Distance-Time Graphs

Exam code: 0625 & 0972

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Written by: Leander Oates
Reviewed by: Caroline Carroll
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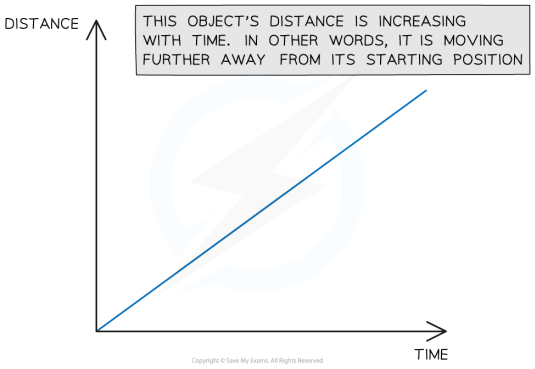
Distance-time graphs

- A **distance-time graph** is used to describe the motion of an object and calculate its speed

Distance-time graph of an object moving at a constant speed



The graph shows a moving object moving further away from its origin at a constant speed

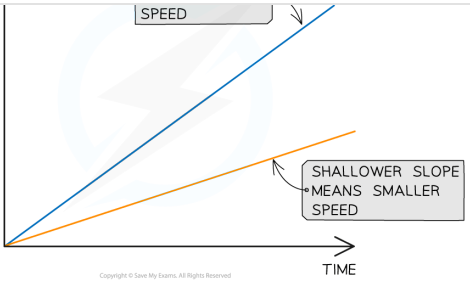
Constant speed on a distance-time graph

- If an object is moving at a **constant speed**, the distance-time graph will be a **straight line**
  - If the constant speed is **zero**, the line will be **horizontal**
  - If the constant speed is **non-zero**, the line will have a **gradient**
- If an object has a speed of zero, the object is **stationary**
  - The distance moved by the object over time is zero
- The **gradient** of a distance-time graph represents the **magnitude** of the object's **velocity**, or its **speed**
  - A **steeper slope**, or a higher gradient, represents a **greater speed**
  - A **shallower slope**, or a lower gradient, represents a **slower speed**

Different speeds on a distance-time graph

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

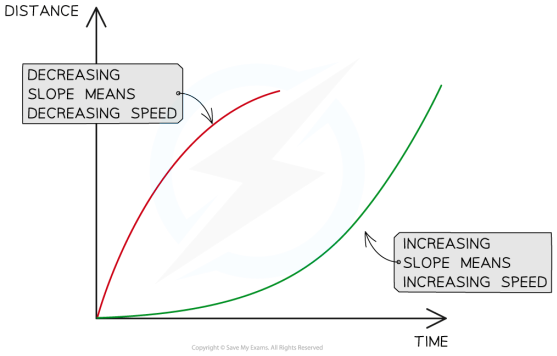


Both of these objects are moving at a constant speed, because the lines are straight. The steeper slope represents the faster speed and the shallower line represents the slower speed.

Changing speed on a distance-time graph

- Often, the speed of an object is **not constant**
- If the speed of an object is **changing**, the object is **accelerating**
- If an object is **accelerating**, the distance-time graph will be a **curved line**
- A **curve** on a distance-time graph is a **changing gradient**
  - If the **gradient increases** over time, the **speed is increasing** over time
  - If the **gradient decreases** over time, the **speed is decreasing** over time

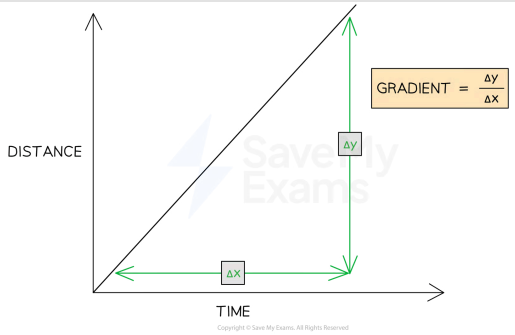
Speed of an object increasing and decreasing on a distance-time graph



Changing speeds are represented by changing slopes, or gradients. The red line shows a decreasing gradient and represents an object slowing down, or decelerating. The green line shows an increasing gradient and represents an object speeding up, or accelerating.

Using distance-time graphs

- The **speed** of a moving object can be calculated from the **gradient** of the line on a **distance-time** graph:

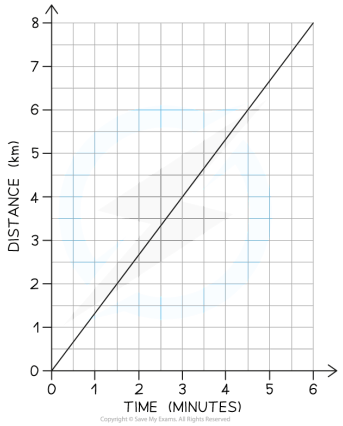


The speed of an object can be found by calculating the gradient of a distance-time graph

- Δy is the **change** in y (distance) values
- Δx is the **change** in x (time) values

Worked Example

A distance-time graph is drawn below for part of a train journey. The train is travelling at a constant speed.



Calculate the speed of the train.

Answer:

Step 1: Draw a large gradient triangle on the graph

- The image below shows a large **gradient triangle** drawn with dashed lines
- Δy and Δx are labelled, using the **units** as stated on each axes

Step 2: Convert units for distance and time into standard units

- The distance travelled, $s = 8\text{ km} \times 1000\text{ m} = 8000\text{ m}$
- The time taken, $t = 6\text{ min} \times 60\text{ s} = 360\text{ s}$

Step 3: State that speed is equal to the gradient of a distance-time graph

- The **gradient** of a **distance-time** graph is equal to the **speed** of a moving object:

$$\text{gradient} = v = \frac{\Delta y}{\Delta x} = \frac{s}{t}$$

Step 4: Substitute values to calculate the speed

$$v = \frac{8000}{360}$$

$$v = 22.2\text{ m/s}$$

Worked Example

A student decides to take a stroll to the park. They find a bench in a quiet spot, take a seat, and read a book on black holes. After some time reading, the student realises they lost track of time and runs home.

A distance-time graph for the trip is drawn below.

home?

(c) What is the total distance travelled by the student?

Answer:

Part (a)

- The student spends **40 minutes** reading his book
- The **flat** section of the line (section B) represents an object which is **stationary**, so section B represents the student sitting on the bench reading
- This section lasts for **40 minutes**

Part (b)

- **Section C** represents the student running home
- The **slope** of the line in section C is **steeper** than the slope in section A
- This means the student was moving at a **faster** speed (running) in section C

Part (c)

- The total distance travelled by the student is **0.6 km**
- The total **distance** travelled by an object is given by the final point on the line; in this case, the line ends at **0.6 km** on the **distance** axis

possible. Examiners tend to award credit if they see a **large gradient triangle** used, so you need to actually draw the lines directly on the graph itself!

Remember to check the **units** on each axis. These may not always be in standard units; in our example, the unit of distance was **km** and the unit of time was **minutes**. Double-check which units to use in your answer.

You could also be asked to plot a distance time graph from data given to you in a question, or be asked to describe the motion of an object from the data given.

You can read more about the use of graphs in exams in the article Graph skills for GCSE Physics

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